EXCEL MASTER INDEX

Basic to Advanced | English & Roman Hindi

ADVANCED EXCEL FORMULAS

FILTER, SORT, UNIQUE, SEQUENCE, LET, LAMBDA, MAP, REDUCE, SCAN, BYROW, BYCOL, MAKEARRAY, GROUPBY, PIVOTBY | Bilingual Guide

Created & Developed by

UTKARSH BAJPAI

utkarshbajpai068@gmail.com

Premium Bilingual Resource | For Students, MIS Executives, Data Analysts & Office Professionals

TABLE OF CONTENTS

1. FILTER
2. SORT
3. SORTBY
4. UNIQUE
5. SEQUENCE
6. RANDARRAY
7. TAKE
8. DROP
9. CHOOSECOLS
10. CHOOSEROWS
11. VSTACK
12. HSTACK
13. LET
14. LAMBDA
15. MAP
16. REDUCE
17. SCAN
18. BYROW
19. BYCOL
20. MAKEARRAY
21. GROUPBY
22. PIVOTBY

UTKARSH BAJPAI

CHAPTER 1: FILTER

PURPOSE - ENGLISH

Filters a range based on conditions.

PURPOSE - ROMAN HINDI

Conditions ke mutabik range ko filter karta hai.

SYNTAX

```
=FILTER(array, include, [if_empty])
```

SYNTAX EXPLANATION

FILTER returns every row that meets the condition.

Roman Hindi: FILTER un saari rows ko deta hai jo condition meet karti hain.

SAMPLE DATASET

Cells: A1:C5 | Values: ID,Name,City
101,Rahul,Delhi
102,Priya,Mumbai
103,Arjun,Delhi
104,Meera,Pune

ACTUAL FORMULA

```
=FILTER(A1:C5, C1:C5="Delhi")
```

EXPECTED OUTPUT

{101,Rahul,Delhi;103,Arjun,Delhi}

STEP-BY-STEP - ENGLISH

- Add the source array.
- Add an include condition with the same row count.
- Press Enter and let the result spill.

STEP-BY-STEP - ROMAN HINDI

- Source array dein.
- Same row count wali include condition dein.
- Enter dabao aur result spill hone do.

REAL OFFICE EXAMPLE

Used for live dashboards that show only one city or status.

Roman Hindi: Aise live dashboards ke liye use hota hai jisme sirf ek city ya status dikhana ho.

COMMON MISTAKES

UTKARSH BAJPAI

- Include range must line up with the array rows.
- Old Excel versions do not support spilling.

IMPORTANT TIPS

- Add an if_empty value such as "No records" for a clean report.

EXCEL VERSION COMPATIBILITY

Excel 2021 / Microsoft 365

CHAPTER 2: SORT

PURPOSE - ENGLISH

Sorts a range by a chosen column.

PURPOSE - ROMAN HINDI

Range ko chuni hui column ke mutabik sort karta hai.

SYNTAX

```
=SORT(array, [sort_index], [sort_order], [by_col])
```

SYNTAX EXPLANATION

SORT returns a sorted copy of the array.

Roman Hindi: SORT array ki sorted copy deta hai.

SAMPLE DATASET

Cells: A2:C5 | Values: 101,Rahul,Delhi
102,Priya,Mumbai
103,Arjun,Delhi
104,Meera,Pune

ACTUAL FORMULA

```
=SORT(A2:C5, 1, -1)
```

EXPECTED OUTPUT

{104,Meera,Pune;103,Arjun,Delhi;102,Priya,Mumbai;101,Rahul,Delhi}

STEP-BY-STEP - ENGLISH

- Add the array.
- Choose the sort column number.
- Use -1 for descending order.

STEP-BY-STEP - ROMAN HINDI

- Array dein.

UTKARSH BAJPAI

- Sort column ka number chuno.
- Descending order ke liye -1 dein.

REAL OFFICE EXAMPLE

Used to show the newest or highest employee IDs first.

Roman Hindi: Sabse naye ya sabse bade employee IDs pehle dikhane ke liye use hota hai.

COMMON MISTAKES

- sort_index is the column inside the array, not the worksheet column letter.

IMPORTANT TIPS

- Use 1 for ascending and -1 for descending.

EXCEL VERSION COMPATIBILITY

Excel 2021 / Microsoft 365

CHAPTER 3: SORTBY

UTKARSH BAJPAI

PURPOSE - ENGLISH

Sorts an array using another range as the key.

PURPOSE - ROMAN HINDI

Kisi doosri range ko key bana kar array sort karta hai.

SYNTAX

```
=SORTBY(array, by_array1, [sort_order1], ...)
```

SYNTAX EXPLANATION

SORTBY keeps the original rows together while sorting by a separate key.

Roman Hindi: SORTBY original rows ko saath rakhta hai aur alag key se sort karta hai.

SAMPLE DATASET

Cells: A2:C5 | Values: 101,Rahul,Delhi
102,Priya,Mumbai
103,Arjun,Delhi
104,Meera,Pune

ACTUAL FORMULA

```
=SORTBY(A2:C5, B2:B5, 1)
```

EXPECTED OUTPUT

{103,Arjun,Delhi;104,Meera,Pune;102,Priya,Mumbai;101,Rahul,Delhi}

STEP-BY-STEP - ENGLISH

- Add the array to sort.
- Add the key range with the same row count.
- Use 1 for ascending.

STEP-BY-STEP - ROMAN HINDI

- Sort karne wala array dein.
- Same row count wali key range dein.
- Ascending ke liye 1 use karo.

REAL OFFICE EXAMPLE

Used to sort an employee table alphabetically by name.

Roman Hindi: Employee table ko naam ke hisaab se alphabetical sort karne ke liye use hota hai.

COMMON MISTAKES

- The key range must have the same row count as the array.
- Sort order values must be 1 or -1.

IMPORTANT TIPS

- You can add more than one sort key.

EXCEL VERSION COMPATIBILITY

Excel 2021 / Microsoft 365

CHAPTER 4: UNIQUE

PURPOSE - ENGLISH

Returns unique values from a range.

PURPOSE - ROMAN HINDI

Range se unique values nikalta hai.

SYNTAX

```
=UNIQUE(array, [by_col], [exactly_once])
```

SYNTAX EXPLANATION

UNIQUE removes duplicate values and spills the remaining list.

Roman Hindi: UNIQUE duplicate values hata kar bachi list spill karta hai.

SAMPLE DATASET

UTKARSH BAJPAI

Cells: A1:A5 | Values: Rahul
Priya
Rahul
Arjun
Priya

ACTUAL FORMULA

```
=UNIQUE(A1:A5)
```

EXPECTED OUTPUT

{Rahul;Priya;Arjun}

STEP-BY-STEP - ENGLISH

- Pass the range.
- Use =UNIQUE(A1:A5).
- Press Enter and let the unique list spill.

STEP-BY-STEP - ROMAN HINDI

- Range pass karo.
- =UNIQUE(A1:A5) use karo.
- Enter dabao aur unique list spill hone do.

REAL OFFICE EXAMPLE

Used to build clean employee, city or product lists.

Roman Hindi: Saaf employee, city ya product lists banane ke liye use hota hai.

COMMON MISTAKES

- Result cells must be empty or the spill shows #SPILL!.

IMPORTANT TIPS

- Use exactly_once TRUE to keep values that appear only once.

EXCEL VERSION COMPATIBILITY

Excel 2021 / Microsoft 365

CHAPTER 5: SEQUENCE

PURPOSE - ENGLISH

Returns a list of sequential numbers.

PURPOSE - ROMAN HINDI

Sequential numbers ki list deta hai.

SYNTAX

UTKARSH BAJPAI

```
=SEQUENCE(rows, [columns], [start], [step])
```

SYNTAX EXPLANATION

SEQUENCE generates a grid of numbers from a start value and step.

Roman Hindi: SEQUENCE start value aur step se numbers ka grid banata hai.

SAMPLE DATASET

Cells: A1:A4 | Values: 3
1
5
5

ACTUAL FORMULA

```
=SEQUENCE(A1, A2, A3, A4)
```

EXPECTED OUTPUT

{5;10;15}

STEP-BY-STEP - ENGLISH

- Add rows and columns.
- Add start and step values.
- Press Enter to spill the sequence.

STEP-BY-STEP - ROMAN HINDI

- Rows aur columns dein.
- Start aur step values dein.
- Enter dabao aur sequence spill hone do.

REAL OFFICE EXAMPLE

Used for automatic invoice numbers, IDs and month numbers.

Roman Hindi: Automatic invoice numbers, IDs aur month numbers ke liye use hota hai.

COMMON MISTAKES

- Rows and columns must be positive numbers.

IMPORTANT TIPS

- Use a negative step to count backwards.

EXCEL VERSION COMPATIBILITY

Excel 2021 / Microsoft 365

CHAPTER 6: RANDARRAY

UTKARSH BAJPAI

PURPOSE - ENGLISH

Returns an array of random numbers.

PURPOSE - ROMAN HINDI

Random numbers ka array deta hai.

SYNTAX

```
=RANDARRAY([rows], [columns], [min], [max], [whole_number])
```

SYNTAX EXPLANATION

RANDARRAY fills a grid with random values inside a minimum and maximum.

Roman Hindi: RANDARRAY grid ko minimum aur maximum ke beech random values se bharta hai.

SAMPLE DATASET

Cells: No cell input | Values: Generated automatically

ACTUAL FORMULA

```
=RANDARRAY(3, 2, 1, 10, TRUE)
```

EXPECTED OUTPUT

Random 3×2 whole-number array

STEP-BY-STEP - ENGLISH

- Choose rows and columns.
- Set the minimum and maximum.
- Use TRUE for whole numbers.

STEP-BY-STEP - ROMAN HINDI

- Rows aur columns chuno.
- Minimum aur maximum set karo.
- Whole numbers ke liye TRUE use karo.

REAL OFFICE EXAMPLE

Used for sample data, simulations and random audits.

Roman Hindi: Sample data, simulations aur random audits ke liye use hota hai.

COMMON MISTAKES

- The result changes every time the sheet recalculates.

IMPORTANT TIPS

- Copy and paste as values when you need a fixed sample.

EXCEL VERSION COMPATIBILITY

Excel 2021 / Microsoft 365

UTKARSH BAJPAI

CHAPTER 7: TAKE

PURPOSE - ENGLISH

Takes a specified number of rows or columns.

PURPOSE - ROMAN HINDI

Array se tay sankhya me rows ya columns leta hai.

SYNTAX

```
=TAKE(array, rows, [columns])
```

SYNTAX EXPLANATION

TAKE trims an array from the top-left corner.

Roman Hindi: TAKE array ko top-left corner se trim karta hai.

SAMPLE DATASET

Cells: A1:A5 | Values: 5
10
15
20
25

ACTUAL FORMULA

```
=TAKE(A1:A5, 3)
```

EXPECTED OUTPUT

{5;10;15}

STEP-BY-STEP - ENGLISH

- Add the array.
- Enter the number of rows to keep.
- Press Enter to spill the result.

STEP-BY-STEP - ROMAN HINDI

- Array dein.
- Kitni rows rakhni hain wo number dein.
- Enter dabao aur result spill hone do.

REAL OFFICE EXAMPLE

Used for Top 3 or Top 5 lists.

Roman Hindi: Top 3 ya Top 5 lists ke liye use hota hai.

COMMON MISTAKES

UTKARSH BAJPAI

- TAKE only works in Microsoft 365.

IMPORTANT TIPS

- Use a negative row count to take rows from the bottom.

EXCEL VERSION COMPATIBILITY

Microsoft 365 (Microsoft 365 / newer version only)

CHAPTER 8: DROP

PURPOSE - ENGLISH

Drops a specified number of rows or columns.

PURPOSE - ROMAN HINDI

Array se tay sankhya me rows ya columns hataata hai.

SYNTAX

```
=DROP(array, rows, [columns])
```

SYNTAX EXPLANATION

DROP removes rows or columns from the start of an array.

Roman Hindi: DROP array ke shuru se rows ya columns hataata hai.

SAMPLE DATASET

Cells: A1:A5 | Values: 5
10
15
20
25

ACTUAL FORMULA

```
=DROP(A1:A5, 2)
```

EXPECTED OUTPUT

{15;20;25}

STEP-BY-STEP - ENGLISH

- Add the array.
- Enter how many rows to remove.
- Press Enter to spill the remaining rows.

STEP-BY-STEP - ROMAN HINDI

- Array dein.

UTKARSH BAJPAI

- Kitni rows hatani hain wo dein.
- Enter dabao aur baaki rows spill hone do.

REAL OFFICE EXAMPLE

Used to remove header rows from imported data.

Roman Hindi: Imported data se header rows hatane ke liye use hota hai.

COMMON MISTAKES

- DROP only works in Microsoft 365.

IMPORTANT TIPS

- Use a negative count to drop rows from the bottom.

EXCEL VERSION COMPATIBILITY

Microsoft 365 (Microsoft 365 / newer version only)

CHAPTER 9: CHOOSECOLS

PURPOSE - ENGLISH

Returns selected columns from an array.

PURPOSE - ROMAN HINDI

Array se selected columns nikalta hai.

SYNTAX

```
=CHOOSECOLS(array, col_num1, [col_num2], ...)
```

SYNTAX EXPLANATION

CHOOSECOLS rebuilds a table using only the columns you name.

Roman Hindi: CHOOSECOLS sirf chuni hui columns se table dobara banata hai.

SAMPLE DATASET

Cells: A1:C3 | Values: 101,Rahul,Delhi
102,Priya,Mumbai
103,Arjun,Delhi

ACTUAL FORMULA

```
=CHOOSECOLS(A1:C3, 1, 3)
```

EXPECTED OUTPUT

{101,Delhi;102,Mumbai;103,Delhi}

STEP-BY-STEP - ENGLISH

UTKARSH BAJPAI

- Add the source array.
- List the column numbers you want.
- Press Enter to spill the selected columns.

STEP-BY-STEP - ROMAN HINDI

- Source array dein.
- Jo columns chahiye unke numbers dein.
- Enter dabao aur selected columns spill hone do.

REAL OFFICE EXAMPLE

Used to export only ID and city columns for a report.

Roman Hindi: Report ke liye sirf ID aur city columns export karne ke liye use hota hai.

COMMON MISTAKES

- Column numbers refer to positions inside the array.

IMPORTANT TIPS

- Use negative numbers to pick columns from the right.

EXCEL VERSION COMPATIBILITY

Microsoft 365 (Microsoft 365 / newer version only)

CHAPTER 10: CHOOSEROWS

PURPOSE - ENGLISH

Returns selected rows from an array.

PURPOSE - ROMAN HINDI

Array se selected rows nikalta hai.

SYNTAX

```
=CHOOSEROWS(array, row_num1, [row_num2], ...)
```

SYNTAX EXPLANATION

CHOOSEROWS rebuilds an array using only the rows you name.

Roman Hindi: CHOOSEROWS sirf chuni hui rows se array dobara banata hai.

SAMPLE DATASET

Cells: A1:C3 | Values: 101,Rahul,Delhi
102,Priya,Mumbai
103,Arjun,Delhi

ACTUAL FORMULA

```
=CHOOSEROWS(A1:C3, 2)
```

UTKARSH BAJPAI

EXPECTED OUTPUT

{102,Priya,Mumbai}

STEP-BY-STEP - ENGLISH

- Add the source array.
- Enter the row numbers you want.
- Press Enter to spill the selected rows.

STEP-BY-STEP - ROMAN HINDI

- Source array dein.
- Jo rows chahiye unke numbers dein.
- Enter dabao aur selected rows spill hone do.

REAL OFFICE EXAMPLE

Used to pull a single employee or the top few records.

Roman Hindi: Ek employee ya top few records nikalne ke liye use hota hai.

COMMON MISTAKES

- Row numbers refer to positions inside the array.

IMPORTANT TIPS

- Combine with SORT to return the first N rows.

EXCEL VERSION COMPATIBILITY

Microsoft 365 (Microsoft 365 / newer version only)

CHAPTER 11: VSTACK

PURPOSE - ENGLISH

Stacks arrays vertically.

PURPOSE - ROMAN HINDI

Arrays ko upar-neeche stack karta hai.

SYNTAX

```
=VSTACK(array1, [array2], ...)
```

SYNTAX EXPLANATION

VSTACK joins arrays one below another.

Roman Hindi: VSTACK arrays ko ek ke neeche jodta hai.

SAMPLE DATASET

Cells: A1:B2 | Values: 1,3
2,4

UTKARSH BAJPAI

ACTUAL FORMULA

```
=VSTACK(A1:A2, B1:B2)
```

EXPECTED OUTPUT

{1;2;3;4}

STEP-BY-STEP - ENGLISH

- Add the first array.
- Add the next array.
- Press Enter to stack them vertically.

STEP-BY-STEP - ROMAN HINDI

- Pehla array dein.
- Agla array dein.
- Enter dabao aur unhe vertically stack karo.

REAL OFFICE EXAMPLE

Used to combine regional lists into one master list.

Roman Hindi: Regional lists ko ek master list me jodne ke liye use hota hai.

COMMON MISTAKES

- Columns that do not exist in one array are padded with #N/A.

IMPORTANT TIPS

- Stack monthly tables and wrap with FILTER for a clean report.

EXCEL VERSION COMPATIBILITY

Microsoft 365 (Microsoft 365 / newer version only)

CHAPTER 12: HSTACK

PURPOSE - ENGLISH

Stacks arrays horizontally.

PURPOSE - ROMAN HINDI

Arrays ko side by side stack karta hai.

SYNTAX

```
=HSTACK(array1, [array2], ...)
```

SYNTAX EXPLANATION

HSTACK joins arrays from left to right.

UTKARSH BAJPAI

Roman Hindi: HSTACK arrays ko left se right jodta hai.

SAMPLE DATASET

Cells: A1:B2 | Values: 1,3
2,4

ACTUAL FORMULA

```
=HSTACK(A1:A2, B1:B2)
```

EXPECTED OUTPUT

{1,3;2,4}

STEP-BY-STEP - ENGLISH

- Add the first array.
- Add the next array.
- Press Enter to place them side by side.

STEP-BY-STEP - ROMAN HINDI

- Pehla array dein.
- Agla array dein.
- Enter dabao aur unhe side by side rakho.

REAL OFFICE EXAMPLE

Used to place calculations beside an imported table.

Roman Hindi: Imported table ke bagal me calculations rakhne ke liye use hota hai.

COMMON MISTAKES

- Rows that do not exist in one array are padded with #N/A.

IMPORTANT TIPS

- Great for adding a calculated ID column next to raw data.

EXCEL VERSION COMPATIBILITY

Microsoft 365 (Microsoft 365 / newer version only)

CHAPTER 13: LET

PURPOSE - ENGLISH

Assigns names to calculation results so a formula is easier to read and faster to calculate.

PURPOSE - ROMAN HINDI

Calculation ke results ko naam deta hai taaki formula padhna aasan ho aur calculation fast ho.

SYNTAX

UTKARSH BAJPAI

```
=LET(name1, value1, [name2, value2], ..., calculation)
```

SYNTAX EXPLANATION

LET stores each value in a name, then uses those names in the final calculation. The last argument is always the result.

Roman Hindi: LET har value ko ek naam me store karta hai aur phir final calculation me un naamo ka use karta hai. Aakhri argument hi result hota hai.

SAMPLE DATASET

Cells: A1:B2 | Values: Price,Qty
100,3

ACTUAL FORMULA

```
=LET(price, A2, qty, B2, price*qty)
```

EXPECTED OUTPUT

300

STEP-BY-STEP - ENGLISH

- Give each value a short name.
- Write the calculation using those names.
- Press Enter to see the single result.

STEP-BY-STEP - ROMAN HINDI

- Har value ko chhota naam do.
- Un naamo se calculation likho.
- Enter dabao aur single result dekho.

REAL OFFICE EXAMPLE

Used in long pricing, tax or incentive formulas so the same expression is not repeated many times.

Roman Hindi: Lambe pricing, tax ya incentive formulas me use hota hai taaki same expression baar-baar na likhna pade.

COMMON MISTAKES

- The last argument must be the calculation, not another name.
- Only Microsoft 365 and Excel 2021+ support LET.

IMPORTANT TIPS

- Use LET to name a VLOOKUP result once and reuse it several times.

EXCEL VERSION COMPATIBILITY

Microsoft 365 / Excel 2021 (Microsoft 365 / newer version only)

CHAPTER 14: LAMBDA

UTKARSH BAJPAI

PURPOSE - ENGLISH

Creates a reusable custom function without VBA or macros.

PURPOSE - ROMAN HINDI

VBA ya macro ke bina reusable custom function banata hai.

SYNTAX

```
=LAMBDA([parameter1, parameter2, ...], calculation)
```

SYNTAX EXPLANATION

LAMBDA defines a function. It can be called directly with brackets, or saved in Name Manager for reuse anywhere.

Roman Hindi: LAMBDA ek function define karta hai. Ise brackets ke saath direct call kar sakte hain, ya Name Manager me save karke kahin bhi use kar sakte hain.

SAMPLE DATASET

Cells: A2 | Values: 100

ACTUAL FORMULA

```
=LAMBDA(x, x*1.18)(A2)
```

EXPECTED OUTPUT

118

STEP-BY-STEP - ENGLISH

- List the parameters after the equals sign.
- Write the calculation using those parameters.
- Call the lambda by adding the argument in brackets.

STEP-BY-STEP - ROMAN HINDI

- Equals sign ke baad parameters likho.
- Un parameters se calculation likho.
- Brackets me argument dekar lambda ko call karo.

REAL OFFICE EXAMPLE

Used to build one company-wide GST or discount function and reuse it in every sheet.

Roman Hindi: Ek company-wide GST ya discount function bana kar har sheet me use karne ke liye use hota hai.

COMMON MISTAKES

- Forgetting the final brackets leaves the function uncalled.
- LAMBDA is not supported before Excel 2021.

IMPORTANT TIPS

- Save a tested LAMBDA in Name Manager and give it a friendly name like GST.

EXCEL VERSION COMPATIBILITY

UTKARSH BAJPAI

CHAPTER 15: MAP

PURPOSE - ENGLISH

Applies a LAMBDA to every value of one or more arrays and returns the mapped array.

PURPOSE - ROMAN HINDI

Ek ya zyada arrays ki har value par LAMBDA apply karta hai aur mapped array deta hai.

SYNTAX

```
=MAP(array1, lambda, [array2], ...)
```

SYNTAX EXPLANATION

MAP walks through the array one value at a time and returns the same size of results.

Roman Hindi: MAP array ki ek-ek value par chalta hai aur same size ke results deta hai.

SAMPLE DATASET

Cells: A2:A4 | Values: 100
200
300

ACTUAL FORMULA

```
=MAP(A2:A4, LAMBDA(x, x*1.18))
```

EXPECTED OUTPUT

{118;236;354}

STEP-BY-STEP - ENGLISH

- Pass the range you want to transform.
- Write a LAMBDA with one parameter.
- Press Enter and the results spill down.

STEP-BY-STEP - ROMAN HINDI

- Jis range ko transform karna hai wo pass karo.
- Ek parameter wala LAMBDA likho.
- Enter dabao aur results neeche spill ho jayenge.

REAL OFFICE EXAMPLE

Used to add GST to a whole price column without filling a formula down.

Roman Hindi: Poori price column me GST add karne ke liye use hota hai, formula neeche fill karne ki zaroorat nahi.

UTKARSH BAJPAI

COMMON MISTAKES

- All arrays must have the same size.
- The last argument must be a LAMBDA, not a plain expression.

IMPORTANT TIPS

- MAP is perfect when you need an array result but your logic is not a built-in function.

EXCEL VERSION COMPATIBILITY

Microsoft 365 / Excel 2021 (Microsoft 365 / newer version only)

CHAPTER 16: REDUCE

PURPOSE - ENGLISH

Reduces an array to one accumulated value using a LAMBDA.

PURPOSE - ROMAN HINDI

LAMBDA ke through array ko ek accumulated value me reduce karta hai.

SYNTAX

```
=REDUCE(initial_value, array, lambda)
```

SYNTAX EXPLANATION

REDUCE starts with the initial value and updates the accumulator once for every item in the array.

Roman Hindi: REDUCE initial value se shuru hota hai aur array ke har item par accumulator update karta hai.

SAMPLE DATASET

Cells: A2:A4 | Values: 100
200
300

ACTUAL FORMULA

```
=REDUCE(0, A2:A4, LAMBDA(acc, val, acc+val))
```

EXPECTED OUTPUT

600

STEP-BY-STEP - ENGLISH

- Choose the starting value, usually 0.
- Pass the array to process.
- Write LAMBDA(acc, val, expression) and press Enter.

STEP-BY-STEP - ROMAN HINDI

UTKARSH BAJPAI

- Starting value chuno, aam taur par 0.
- Process karne wali array pass karo.
- LAMBDA(acc, val, expression) likho aur Enter dabao.

REAL OFFICE EXAMPLE

Used for custom totals, running balances, and complex conditional sums that SUMIFS cannot handle.

Roman Hindi: Custom totals, running balances aur un complex conditional sums ke liye use hota hai jo SUMIFS se nahi hote.

COMMON MISTAKES

- The LAMBDA must have exactly two parameters: accumulator and value.
- Do not put the whole array inside the LAMBDA body.

IMPORTANT TIPS

- Return a concatenated text list by starting with \"\" as the initial value.

EXCEL VERSION COMPATIBILITY

Microsoft 365 / Excel 2021 (Microsoft 365 / newer version only)

CHAPTER 17: SCAN

PURPOSE - ENGLISH

Returns a running total or cumulative result for each value in an array.

PURPOSE - ROMAN HINDI

Array ki har value ke liye running total ya cumulative result deta hai.

SYNTAX

```
=SCAN(initial_value, array, lambda)
```

SYNTAX EXPLANATION

SCAN works like REDUCE but keeps every intermediate accumulator value, producing a spilled array.

Roman Hindi: SCAN REDUCE jaisa hai lekin har intermediate accumulator value rakhta hai, isliye spilled array deta hai.

SAMPLE DATASET

Cells: A2:A4 | Values: 100
200
300

ACTUAL FORMULA

```
=SCAN(0, A2:A4, LAMBDA(acc, val, acc+val))
```

UTKARSH BAJPAI

EXPECTED OUTPUT

{0;100;300;600}

STEP-BY-STEP - ENGLISH

- Set the initial value, usually 0.
- Pass the array to scan.
- Write LAMBDA(acc, val, acc+val) and press Enter.

STEP-BY-STEP - ROMAN HINDI

- Initial value set karo, aam taur par 0.
- Scan karne wali array pass karo.
- LAMBDA(acc, val, acc+val) likho aur Enter dabao.

REAL OFFICE EXAMPLE

Used for running balances, cumulative sales, and month-to-date reports.

Roman Hindi: Running balances, cumulative sales aur month-to-date reports ke liye use hota hai.

COMMON MISTAKES

- The result has one more row than the input because the initial value is included.
- Use REDUCE instead when you only need the final total.

IMPORTANT TIPS

- SCAN is the formula version of a running total helper column.

EXCEL VERSION COMPATIBILITY

Microsoft 365 / Excel 2021 (Microsoft 365 / newer version only)

CHAPTER 18: BYROW

PURPOSE - ENGLISH

Applies a LAMBDA to each row of an array and returns one result per row.

PURPOSE - ROMAN HINDI

Array ki har row par LAMBDA apply karta hai aur har row ke liye ek result deta hai.

SYNTAX

```
=BYROW(array, lambda)
```

SYNTAX EXPLANATION

BYROW treats each row as a mini-array, so functions like SUM or MAX can work row by row.

Roman Hindi: BYROW har row ko chhoti array maanta hai, isliye SUM ya MAX jaisi functions row by row kaam karti hain.

UTKARSH BAJPAI

SAMPLE DATASET

Cells: A1:B3 | Values: 10,20
30,40
50,60

ACTUAL FORMULA

```
=BYROW(A1:B3, LAMBDA(r, SUM(r)))
```

EXPECTED OUTPUT

{30;70;110}

STEP-BY-STEP - ENGLISH

- Pass the two-dimensional array.
- Write LAMBDA(r, SUM(r)).
- Press Enter to get one total per row.

STEP-BY-STEP - ROMAN HINDI

- Two-dimensional array pass karo.
- LAMBDA(r, SUM(r)) likho.
- Enter dabao aur har row ka total pao.

REAL OFFICE EXAMPLE

Used to total each employee's monthly figures without a helper column.

Roman Hindi: Har employee ke monthly figures ka total helper column ke bina nikalne ke liye use hota hai.

COMMON MISTAKES

- The LAMBDA must accept exactly one parameter.
- The input must be a range or array, not a single value.

IMPORTANT TIPS

- Use BYROW with MAX to find each row's highest month.

EXCEL VERSION COMPATIBILITY

Microsoft 365 / Excel 2021 (Microsoft 365 / newer version only)

CHAPTER 19: BYCOL

PURPOSE - ENGLISH

Applies a LAMBDA to each column of an array and returns one result per column.

PURPOSE - ROMAN HINDI

Array ke har column par LAMBDA apply karta hai aur har column ke liye ek result deta hai.

SYNTAX

UTKARSH BAJPAI

```
=BYCOL(array, lambda)
```

SYNTAX EXPLANATION

BYCOL treats each column as a mini-array and returns a horizontal spilled result.

Roman Hindi: BYCOL har column ko chhoti array maanta hai aur horizontal spilled result deta hai.

SAMPLE DATASET

Cells: A1:B3 | Values: 10,20
30,40
50,60

ACTUAL FORMULA

```
=BYCOL(A1:B3, LAMBDA(c, SUM(c)))
```

EXPECTED OUTPUT

{90,120}

STEP-BY-STEP - ENGLISH

- Pass the two-dimensional array.
- Write LAMBDA(c, SUM(c)).
- Press Enter to get one total per column.

STEP-BY-STEP - ROMAN HINDI

- Two-dimensional array pass karo.
- LAMBDA(c, SUM(c)) likho.
- Enter dabao aur har column ka total pao.

REAL OFFICE EXAMPLE

Used to total each month's column across all employees.

Roman Hindi: Saare employees ke across har month ka column total nikalne ke liye use hota hai.

COMMON MISTAKES

- The LAMBDA must accept exactly one parameter.
- The result spills across columns, so keep cells to the right empty.

IMPORTANT TIPS

- Pair BYCOL with AVERAGE to build a compact KPI row.

EXCEL VERSION COMPATIBILITY

Microsoft 365 / Excel 2021 (Microsoft 365 / newer version only)

CHAPTER 20: MAKEARRAY

UTKARSH BAJPAI

PURPOSE - ENGLISH

Builds an array of any size by running a LAMBDA for every row and column position.

PURPOSE - ROMAN HINDI

Har row aur column position par LAMBDA chala kar kisi bhi size ka array banata hai.

SYNTAX

```
=MAKEARRAY(rows, columns, lambda)
```

SYNTAX EXPLANATION

MAKEARRAY passes the row number and column number into the LAMBDA and fills each cell with the returned value.

Roman Hindi: MAKEARRAY row number aur column number LAMBDA me deta hai aur har cell ko returned value se bharta hai.

SAMPLE DATASET

Cells: A1:B2 | Values: 2,3
4,5

ACTUAL FORMULA

```
=MAKEARRAY(2, 2, LAMBDA(r, c, r*c))
```

EXPECTED OUTPUT

{1,2;2,4}

STEP-BY-STEP - ENGLISH

- Set the number of rows and columns.
- Write LAMBDA(r, c, expression) using the position numbers.
- Press Enter to spill the generated grid.

STEP-BY-STEP - ROMAN HINDI

- Rows aur columns ki sankhya set karo.
- Position numbers use karte hue LAMBDA(r, c, expression) likho.
- Enter dabao aur generated grid spill hone do.

REAL OFFICE EXAMPLE

Used to generate multiplication tables, seat maps, and test data grids.

Roman Hindi: Multiplication tables, seat maps aur test data grids banane ke liye use hota hai.

COMMON MISTAKES

- Rows and columns must be positive whole numbers.
- The LAMBDA must have exactly two parameters.

IMPORTANT TIPS

- Combine MAKEARRAY with RANDARRAY-style logic to create sample datasets.

UTKARSH BAJPAI

EXCEL VERSION COMPATIBILITY

Microsoft 365 / Excel 2021 (Microsoft 365 / newer version only)

CHAPTER 21: GROUPBY

PURPOSE - ENGLISH

Groups rows by one or more fields and calculates a summary without a PivotTable.

PURPOSE - ROMAN HINDI

Ek ya zyada fields se rows group karta hai aur PivotTable ke bina summary calculate karta hai.

SYNTAX

```
=GROUPBY(row_fields, values, function, [field_headers], [total_depth], [sort_order],
[filter_array])
```

SYNTAX EXPLANATION

GROUPBY produces a spilled summary table: one row per unique group and one row for the chosen aggregation.

Roman Hindi: GROUPBY spilled summary table banata hai: har unique group ke liye ek row aur chuni hui aggregation ke liye ek result.

SAMPLE DATASET

Cells: A1:B5 | Values: City,Sales
Delhi,100
Mumbai,150
Delhi,50
Mumbai,200

ACTUAL FORMULA

```
=GROUPBY(A2:A5, B2:B5, SUM)
```

EXPECTED OUTPUT

{Delhi,150;Mumbai,350}

STEP-BY-STEP - ENGLISH

- Select the column that identifies each group.
- Select the values to aggregate.
- Pass the function name, for example SUM, and press Enter.

STEP-BY-STEP - ROMAN HINDI

- Wo column chuno jo har group ko identify kare.
- Aggregate karne wali values chuno.
- Function ka naam do, jaise SUM, aur Enter dabao.

UTKARSH BAJPAI

REAL OFFICE EXAMPLE

Used for quick city-wise, product-wise or salesperson-wise totals inside a formula.

Roman Hindi: Formula ke andar quick city-wise, product-wise ya salesperson-wise totals ke liye use hota hai.

COMMON MISTAKES

- GROUPBY needs Microsoft 365 with the new calculation engine.
- The function argument is written without brackets, for example SUM.

IMPORTANT TIPS

- Add a third grouping column as an array constant to group by multiple fields.

EXCEL VERSION COMPATIBILITY

Microsoft 365 (Microsoft 365 / newer version only)

CHAPTER 22: PIVOTBY

PURPOSE - ENGLISH

Builds a PivotTable-style summary with row fields, column fields and values using one formula.

PURPOSE - ROMAN HINDI

Ek formula se row fields, column fields aur values wali PivotTable-style summary banata hai.

SYNTAX

```
=PIVOTBY(row_fields, col_fields, values, function, [field_headers], [row_total_depth],
[row_sort_order])
```

SYNTAX EXPLANATION

PIVOTBY puts unique values of one field down the rows and another across the columns, then aggregates the values at each intersection.

Roman Hindi: PIVOTBY ek field ki unique values ko rows me aur doosri ko columns me rakhta hai, phir har intersection par values aggregate karta hai.

SAMPLE DATASET

Cells: A1:C5 | Values: City,Product,Sales
Delhi,A,100
Mumbai,A,150
Delhi,B,50
Mumbai,B,200

ACTUAL FORMULA

```
=PIVOTBY(A2:A5, B2:B5, C2:C5, SUM)
```

UTKARSH BAJPAI

EXPECTED OUTPUT

{Delhi,150,50;Mumbai,150,200}

STEP-BY-STEP - ENGLISH

- Choose the row field.
- Choose the column field.
- Add the value field and the SUM function.

STEP-BY-STEP - ROMAN HINDI

- Row field chuno.
- Column field chuno.
- Value field aur SUM function add karo.

REAL OFFICE EXAMPLE

Used for city-by-product sales matrices that update automatically when data changes.

Roman Hindi: City-by-product sales matrices ke liye use hota hai jo data change hone par apne aap update hoti hain.

COMMON MISTAKES

- PIVOTBY is a Microsoft 365-only function.
- The function argument is written as SUM, not SUM().

IMPORTANT TIPS

- Use PIVOTBY when you want a PivotTable that refreshes with the worksheet.

EXCEL VERSION COMPATIBILITY

Microsoft 365 (Microsoft 365 / newer version only)

UTKARSH BAJPAI